

Virtual Streamer with Conversational and Tactile Interaction

Vaishnavi Josyula*

The University of Texas at Dallas

Sowresh Mecheri-Senthil†

The University of Texas at Dallas

Abbas Khawaja‡

The University of Texas at Dallas

Jose M. Garcia§

The University of Texas at Dallas

Ayush Bhardwaj¶

The University of Texas at Dallas

Ashish Pratap||

The University of Texas at Dallas

Jin Ryong Kim**

The University of Texas at Dallas



Figure 1: (a) A virtual environment, (b) an avatar playing a game, and (c) a user interaction with haptic feedback.

ABSTRACT

This paper introduces an approach to an interactive virtual streamer that provides conversational and tactile interactions, offering insights into levels of immersion and personalization for the user. User interaction is categorized into various types, such as the spatial nature of the virtual streamer's stream, along with conversational and tactile interactions with the streamer. We deploy the virtual streamer in a VR environment and conclude the efficiency of user interactions with the virtual streamer via a scenario-based assessment.

Index Terms: Virtual Streamer, Haptic Interaction, Conversational Interaction, Avatar Interaction, Large Language Models

1 INTRODUCTION

Streaming has evolved into a prevalent form of entertainment, offering users a more personal connection with content creators. Virtual streamers, or VTubers, are streamers who use an avatar instead of a face camera to animate their interactions, adding an engaging layer to the streaming experience. Traditionally, virtual streamers are controlled by humans where the avatar's actions, facial expressions, and lip movements are mapped to the human's motions and expressions. Recently, there has been a notable surge in virtual streamers powered by artificial intelligence (AI) who have gained massive popularity in the streaming world. These AI-powered virtual streamers operate autonomously with little to no human intervention to control the avatar's facial expressions, play games, and interact with users [3].

Although AI-powered virtual streamers are intelligent and can communicate with users while streaming games, they pose limitations in terms of user engagement and personalization. Since virtual

streamers stream to a large audience, they can not cater to each user's comments and, therefore, hinder effective user engagement. Furthermore, virtual streamers often face challenges in representing realism due to limitations in current technologies [1]. They operate behind a screen, which hinders users from directly and tactically interacting with virtual streamers, further decreasing user immersion.

We propose incorporating an AI-powered virtual streamer into VR, allowing users to engage in social interactions, thereby adding another layer of realism to the stream. Our proposal utilizes multi-modal interaction, combining visual, auditory, conversational, and tactile interactions with a streaming service to enhance user interaction with the virtual streamer. Within the VR environment, the virtual streamer can engage in personalized one-to-one conversations with the user, creating an immersive streaming experience that is tailored to the user. Moreover, the VR user can interact with the virtual streamer through speech and touch, allowing for direct and effective user interaction, which fosters a dynamic streaming experience and user engagement (see Figure 1).

2 DESIGN

Our approach showcases an interactive virtual streamer in VR to increase immersion and user engagement. The virtual streamer is presented as an avatar to simulate human appearance and communicate with users through conversational and tactile interactions.

2.1 Virtual Environment

The virtual environment is created in Unity, featuring a room with a TV, desk, and chair (see Figure 1(a)). An interactive virtual streamer is presented as an avatar for conversational and full-body haptic interactions. A user can enter the VR scene, watch a game stream on the TV, and interact with the avatar.

2.2 Streaming Service

The game is streamed on Twitch [2] on a server using OBS Studio (see Figure 1(b) and Figure 2). The Twitch server accesses the stream and projects it to the VR environment. We created four games: Chrome Dino, Donkey Kong, Asteroids, and Flappy Bird. A user can watch these games on the TV via Twitch stream. The games can be switched using voice interaction. Each game has a

*e-mail: vaishnavi.josyula@utdallas.edu

†e-mail: sowresh.mecheri-senthil@utdallas.edu

‡e-mail: abbas.khawaja@utdallas.edu

§e-mail: jose.garcia3@utdallas.edu

¶e-mail: ayush.bhardwaj@utdallas.edu

||e-mail: ashish.pratap@utdallas.edu

**e-mail: jin.kim@utdallas.edu

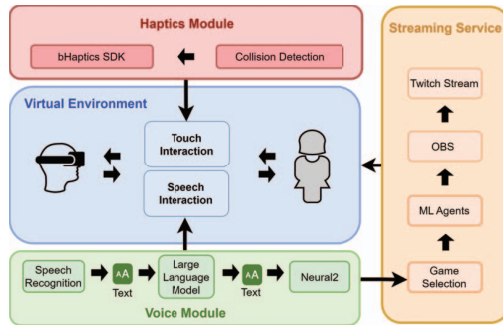


Figure 2: Architecture

machine learning (ML) model trained using ML Agents, allowing the streamer to play each game for the user's entertainment.

2.3 Conversation Module

Our virtual streamer engages in intelligent conversations with the user through a three-step process. It first converts the user's speech to text, then generates a response based on the text, and finally converts the response back to speech (see Figure 2).

Speech to Text: The VR headset's microphone captures the user's speech, while a Python script in the background actively awaits and processes the audio input. The user's spoken words are transcribed into text using Python's 'SpeechRecognition' library.

Response Generation: The transcribed text is sent as input to a large language model (LLM). We used the GPT 3.5 LLM, which can hold conversations with the user in a manner comparable to human conversations.

Text to Speech: After generating a response, the virtual streamer replies to the user with speech. The speech was generated using a Neural2 voice using deep neural networks trained to generate realistic-sounding speech. The generated audio is then played through the VR headset's speakers. Using ULipSync, the virtual streamer's lips sync with the played audio.

2.4 Haptics Module

Our platform supports haptic interaction (see Figure 1(c) and Figure 3) with the user. The avatar can detect when the user is within its proximity and collision. In the occurrence of a collision, the user will receive haptic feedback via a vest (bHaptics TactSuit X40) and gloves (bHaptics TactGlove DK1) based on the type of collision.

Two types of haptic interaction are provided: a handshake and a hug. The avatar detects when the user's hand is close to the avatar's hand, prompting a vibration that simulates a handshake through a glove. The glove uses actuators at the fingertips to produce these vibrations, which occur in a timed sequence. Likewise, the avatar detects when the user's headset is near the avatar's spine, thereby detecting the hug. The vest uses actuators in a sequence to simulate the feeling of having hands around the user's back, with the middle actuators vibrating first and the side actuators vibrating next. Subsequently, the gloves vibrate after the vest, mirroring the conclusive phase of the hug: the feeling of hands encircling the body. We used inverse kinematics to simulate a hug between the avatar and the user.

In addition to the haptic interactions, the avatar waves when greeting the user. The streamer's avatar is rigged using VRoid Studio. We used the open-source library from Mixamo to apply three animations to the rigged avatar: idle, wave, and handshake. The avatar defaults to the idle position, but switches to the wave when greeting the user and to the handshake when it detects the user's hands near the streamer's hands. The avatar's eyes also blink occasionally as a feature.

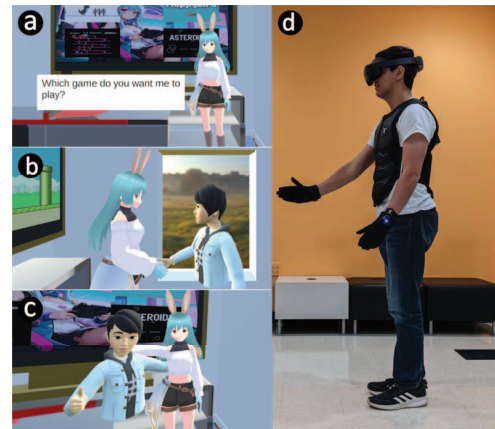


Figure 3: Types of user interactions with the virtual streamer: (a) speech, (b) handshake, (c) hug, (d) a user performing a handshake.

3 AVATAR INTERACTION AND SCENARIOS

In our scene, we provide conversational and tactile interactions with an avatar, offering an immersive and natural game stream experience.

Scenario 1: Conversational interaction Upon entering the VR scene, the user can talk with the avatar. The user can ask the avatar about herself, or topics such as her backstory and favorite hobbies. This allows the user to connect more easily with the avatar, and build personal relationships and trust, engaging the user with the streamed content. The user can ask various questions about the game content and receive appropriate answers. Direct control of streaming content is also allowed, giving a more personalized experience than 2D-based Twitch services.

Scenario 2: Haptics-based interaction Users can enjoy a hug (see Figure 3(c)) and a friendly handshake (see Figure 3(b)), complemented with haptic feedback. This tactile enhancement elevates the quality of the one-to-one virtual experience and cultivates a deeper and more meaningful connection with the avatar.

4 CONCLUSION

This paper presented a novel approach to increase user engagement with virtual streamers in VR through conversational and tactile interactions with a virtual streamer avatar. We aimed to improve emotional connections, streamline the streaming process, and enhance the overall immersive experience for content creators and users. Our prototype showed a holistic approach that incorporates multimodal interactions for the enjoyment of virtual streaming experiences.

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